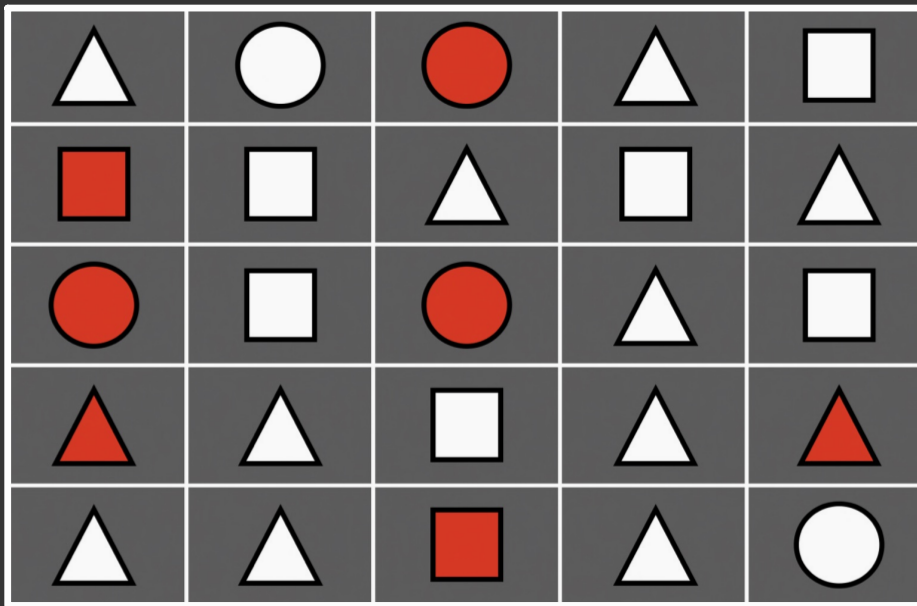


the law of total
probability

CSCI
270

what is the probability of drawing
a **red** shape?



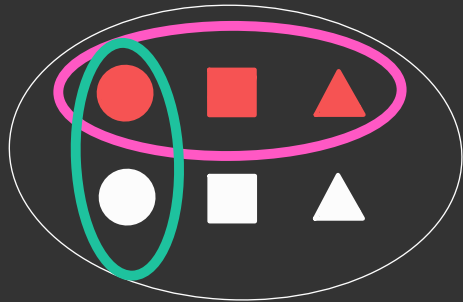
what is the probability of drawing
a **red** shape?

△	○	●	△	□
■	□	△	□	△
●	□	●	△	□
▲	△	□	△	▲
△	△	■	△	○

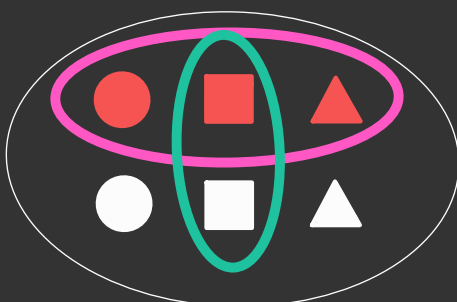
case-by-case
analysis can be helpful

$$P(\text{●}) + P(\text{■}) + P(\text{▲})$$

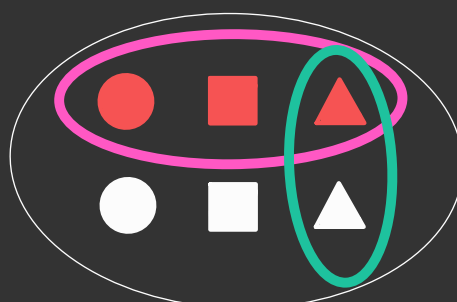
$P(\text{you draw red } \cap \text{ you draw } \text{○})$



$P(\text{you draw red } \cap \text{ you draw } \text{□})$



$P(\text{you draw red } \cap \text{ you draw } \text{▲})$



what is the probability of drawing
a **red** shape?

△	○	●	△	□
■	□	△	□	△
●	□	●	△	□
△	△	□	△	△
△	△	■	△	○

case-by-case

analysis can be helpful

$$P(\bullet) + P(\blacksquare) + P(\blacktriangle)$$

$$\frac{3}{25}$$

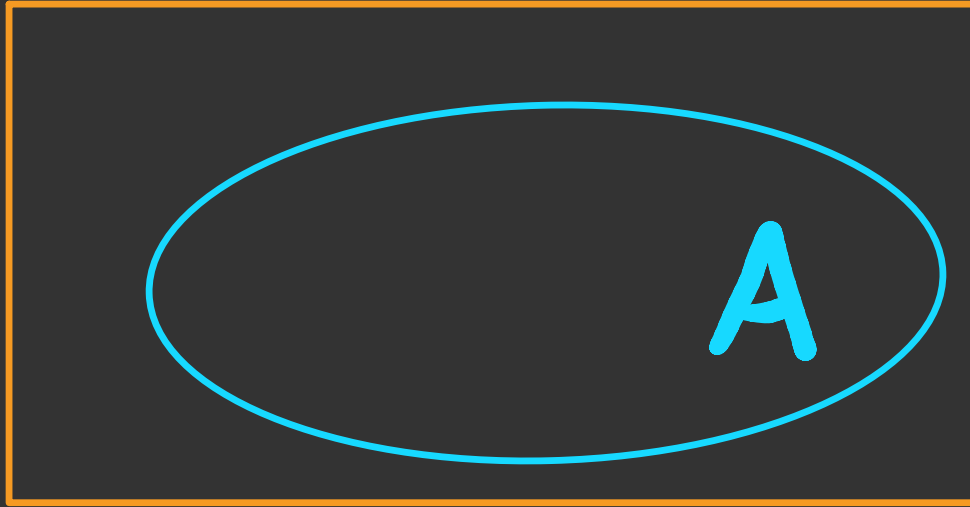
+

$$\frac{2}{25}$$

+

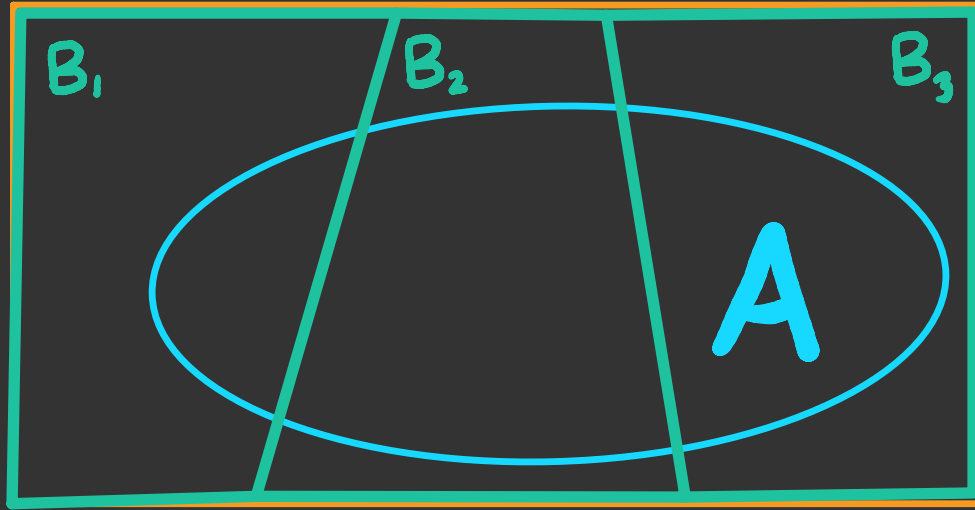
$$\frac{2}{25}$$

what is the probability
of event **A**?



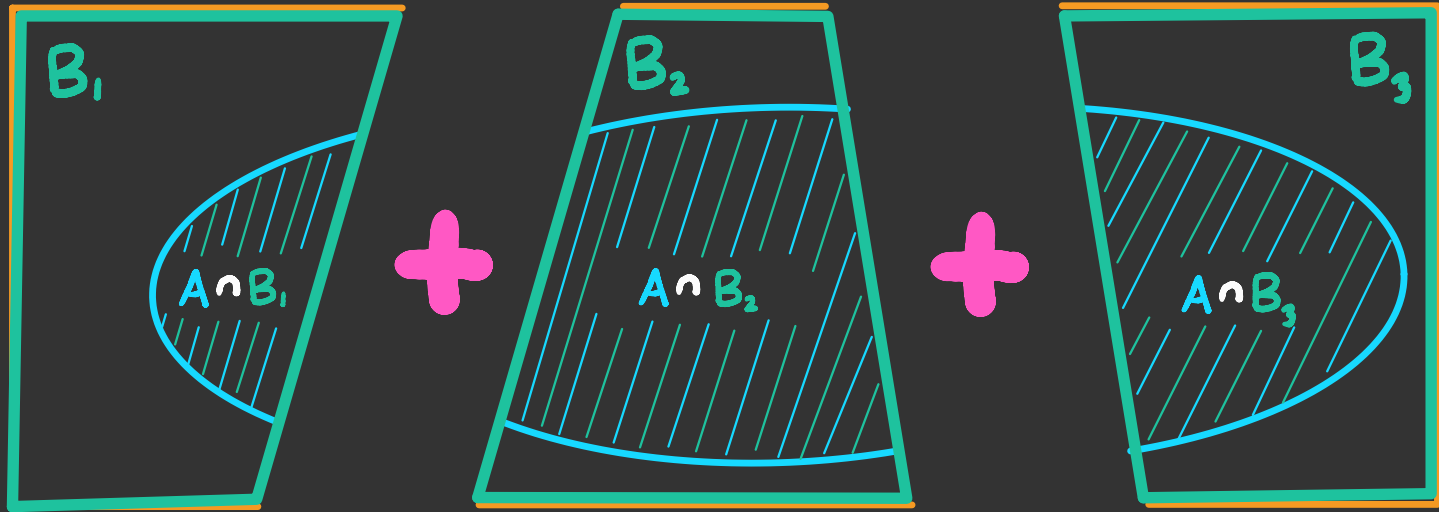
sample space Ω

one strategy: perform a case analysis
over a set of disjoint events $B_1, \dots, B_k$
that cover the sample space



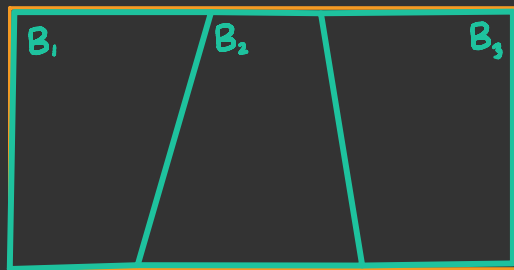
sample space Ω

one strategy: perform a case analysis
over a set of disjoint events $B_1, \dots, B_k$
that cover the **sample space**



the law of total probability

let $B_1, \dots, B_k$ be a set of disjoint events whose union is the entire sample space

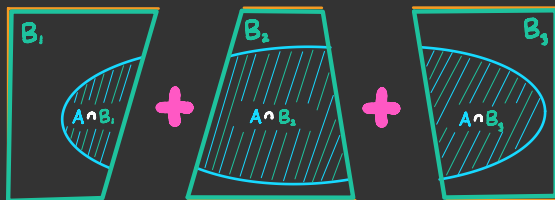


then the probability of any event A



can be computed as follows:

$$P(A) = \sum_{i=1}^k P(A \cap B_i)$$



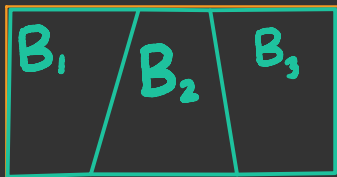
proof of the law of total probability

let A be an event



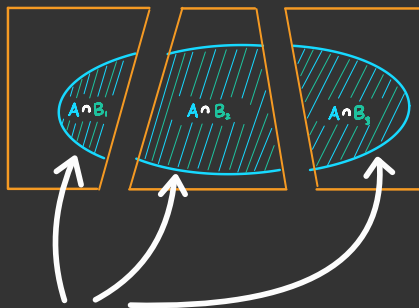
let $B_1, \dots, B_k$ be a set of disjoint events whose union is the entire

sample space



then:

$$A = (A \cap B_1) \cup \dots \cup (A \cap B_k)$$



these are disjoint,
and by definition

if A and B are disjoint events,

$$P(A \cup B) = P(A) + P(B)$$

so:

$$P(A) = \sum_{i=1}^k P(A \cap B_i)$$

useful probability rules

A is an event

conditional probability

$y|x$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

bayes rule



$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

total probability



$$P(A) = \sum_{i=1}^k P(A \cap B_i)$$

B is an event

B₁, ..., B_k are
disjoint events
whose union is the entire
sample space